

Assessment of Heart Rate Variability and Respiratory Sinus Arrhythmia via Doppler Radar

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Abstract—An investigation of heart rate variability (HRV) and respiratory sinus arrhythmia (RSA) indices using data obtained from Doppler radar cardiopulmonary remote sensing is presented in this paper. High accuracy in extracting the HRV and RSA indices was achieved using a direct-conversion quadrature radar system with linear demodulation method. This method was validated using data obtained from 12 human subjects in seated and supine positions. For supine position measurements, all standard deviation of normal beat-to-beat interval indices from Doppler radar and electrocardiogram reference differed less than ± 9 ms of each other, while all the root mean square of differences of successive normal beat-to-beat interval indices differed less than ± 76 ms. The measurements from subjects in seated and supine positions with normal RSA demonstrated that the results from radar correlated well with both respiratory piezoresistor chest belts. Higher level of HRV and RSA was detected in seated position, as expected.

Index Terms—Doppler radar, heart rate variability (HRV), remote physiological sensing, respiratory sinus arrhythmia (RSA).

I. INTRODUCTION

HEART RATE variability (HRV) refers to the beat-to-beat alteration in heart rate. It is regarded as an indicator of the activity of autonomic regulation of circulatory function; reflecting the manner in which the cardiovascular regulatory system responds to demands, stress, and illness [1]. Reduced HRV has been used as a marker of reduced vagal activity. Medical research has associated lowered HRV with aging, decreased autonomic activity, sleep apnea, diabetic neuropathy, and increased risk of sudden cardiac death after acute myocardial infarction (MI) [2]–[6]. Psychological research indicates that depression, panic disorders, and anxiety have negative

impact on autonomic function. Autonomic imbalance caused by significant mental and emotional stress increases risk of acute MI followed by sudden cardiac death [6]. Multiple pharmacological studies suggest that HRV is a useful indicator to quantitatively measure physiological changes during treatment [7], [8]. HRV assessment stands out as one of the most promising methods for the future of general health evaluation.

Variation in heart rate is believed to be a result from: 1) daily (circadian) rhythm, which varies the heart rate over cycles of 5 h or more; 2) temperature regulation, which varies with about 25-s cycles; 3) cardiac sympathetic (fight and flight) nervous activity, which varies with about 6-s cycles; and 4) synchronization with respiratory rhythm, which varies the beat-to-beat intervals with each breath [1]. The natural cycle of arrhythmia that occurs through the influence of breathing on the flow of sympathetic and vagus impulses to the sinoatrial node is called respiratory sinus arrhythmia (RSA). The rhythm fluctuates with the phase of respiration—cardio-acceleration during inspiration, and cardio-deceleration during expiration. RSA is the most commonly investigated component of HRV. While RSA is usually identified as the high-frequency (HF) content in HRV spectrum, an independent measure of respiration is necessary for this assessment.

Continuous beat-to-beat interval or instantaneous heart rate measurements are the source information for HRV analysis. The electrocardiogram (ECG) is traditionally considered the standard way to measure the beat-to-beat intervals. The other approach to measure beat-to-beat intervals is measurement of pulse waves by a photoplethysmograph (PPG) or a piezoresistor sensor, which is less invasive and simpler than an ECG. Nevertheless, all methods mentioned above require patients to be tethered to the sensing devices. Current techniques for measuring respiration effort/movement typically require direct contact with the patient through various chest bands that may impede unrestricted chest motion, or require the use of face masks, which may change the subject's respiration pattern. The ability of Doppler radar to simultaneously obtain both heart rate and respiration measurement remotely could prove to be useful for future HRV and RSA research.

The challenge for using Doppler radar for HRV analysis is whether it could provide acceptable accuracy for a period of time needed (from 2 to 5 min for short-term analysis, and up to 24 h for long-term analysis). The Doppler radar heart signal does not exhibit a peak as sharp as the ECG signal since it is a measure of mechanical motion of the chest due to heart motion. However, our previous efforts have demonstrated that Doppler radar has potential to provide a practical tool for HRV assessment [9]–[11]. Preliminary results have shown that using the

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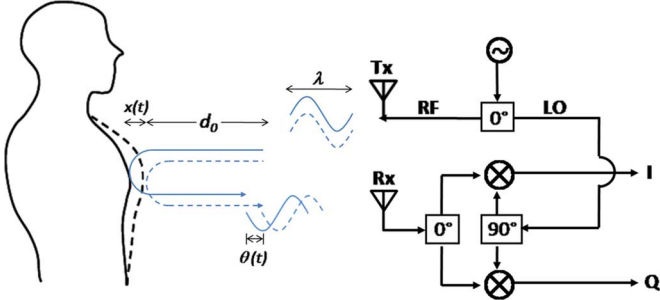


Fig. 1. Block diagram of a quadrature Doppler radar cardiopulmonary measurement system. The local oscillator (LO) signal is divided by a two-way 90° power splitter to get two orthonormal baseband signals (I and Q) to ensure both channels are not in a minimum sensitivity spot.

arctangent demodulation method results in standard deviation of normal beat-to-beat intervals (SDNNs) with an accuracy of about 30 ms for a small sample of seated subjects [10], and 10 ms when using the linear demodulation method [11]. The study presented in this paper validates this approach using data from 12 human subjects in seated and supine positions. Comprehensive HRV and RSA index analysis is presented, demonstrating that those parameters can be obtained with sufficient accuracy for the majority of subjects. For supine position measurements, all SDNNs of Doppler radar and ECG reference differed less than ± 9 ms of each other, while all the root mean square of differences of successive beat-to-beat intervals (RMSSDs) differed less than ± 76 ms. The measurements from subjects in seated and supine positions with normal RSA demonstrated that the results from radar correlated well with both respiratory piezoresistor chest belts. A higher level of HRV and RSA was detected in the supine position, as expected.

The following sections present the background on physiological sensing using Doppler radar, the experimental results from a quadrature receiver system, and the analytical explanation of HRV and RSA assessment using results from Doppler radar.

II. DOPPLER RADAR SYSTEM

The ability to remotely detect vital signs such as heart beat and respiration is particularly useful in situations where direct contact with the subject is either impossible or unwanted. The use of microwave measurement of cardiopulmonary movements in men and animals has been demonstrated as early as the 1970s [12]. A Doppler radar for physiological motion sensing, as shown in Fig. 1, transmits a radio wave signal and receives a motion-modulated signal reflected from a target. The RF wave reflected at a moving surface undergoes a phase shift proportional to the surface displacement. A quadrature receiver is a chosen architecture to overcome the demodulation sensitivity to a target's distance, which occurs in single channel direct conversion systems [15]–[18]. Assuming that the target's time-varying displacement is given by $x(t)$, the baseband amplitude due to receiver and mixer gain is given by A_B , the constant phase shift due to the nominal distance to the target is given by θ , and the residual phase noise is given by $\Delta\phi(t)$, the quadrature baseband outputs can be expressed as

$$B_I(t) = A_B \cos \left(\theta + \frac{\pi}{4} + \frac{4\pi x(t)}{\lambda} + \Delta\phi(t) \right) \quad (1)$$

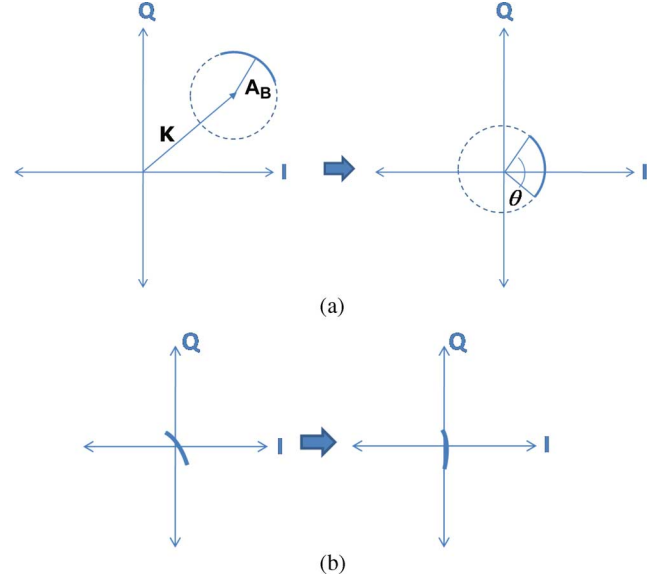


Fig. 2. Nonlinear versus linear demodulations. (a) Arc tangent demodulation (nonlinear); the dc cluster is removed and the center of the arc is identified before the phase information is extracted. (b) Linear demodulation is applied when the arc is small; the arc length is linearly approximated after it is rotated to be parallel to the Q-axis. The signal in (a) contained all the dc information, while (b) is ac coupled.

$$B_Q(t) = A_B \sin \left(\theta + \frac{\pi}{4} + \frac{4\pi x(t)}{\lambda} + \Delta\phi(t) \right). \quad (2)$$

Both arctangent (nonlinear) and linear demodulation methods, illustrated in Fig. 2, can be used to estimate $x(t)$. The phase information could be recovered directly via the arctangent of the baseband signals [10] without needing the small angle approximation. However, if the signals fall into one of these three categories: 1) arc length resulted from the phase change is relatively short; 2) signal-to-noise ratio (SNR) is low; or 3) ac coupling is used, then the linear demodulation method may be a more accurate choice [17]. Linear demodulation is a method of transposing multidimensional data to a single dimension, suppressing redundant information, and maximizing the variance in the data. For the quadrature receiver, first, any dc offset is removed from the data, and the covariance matrix between the in-phase (I) and quadrature (Q) channels is found. The I-Q arc is rotated such that it will be parallel to the Q-axis. To achieve this, the received signal is multiplied by the transpose of the covariance matrix eigenvector. After the rotation, the Q-component is always in an optimum point and the I-component is always in a null point. Thus, this Q-component can be used as the demodulation signal. Mathematical representation of the linear demodulation can be written as

$$\hat{x}[n] = d^T r[n] \quad (3)$$

where $\hat{x}[n]$ is the output of the linear demodulation, d^T is the transpose matrix of the eigenvector from the covariance matrix, and $r[n]$ is the matrix composed of the I and Q received signals. It was demonstrated that the linear demodulation method could extract the heart signals with high accuracy if ac coupling is used [11], [17], and thus was chosen as the demodulation method for

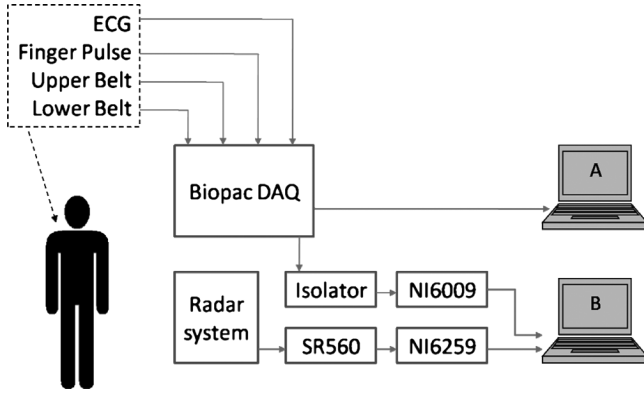


Fig. 3. Block diagram of test protocol setup. The reference signals: ECG finger pulse, upper chest belt, and lower chest belt, were completely isolated from the radar system. The sync signal provided the marker when the Biopac reference system started capturing to line up the data from the two DAQ systems. Computer A recorded the reference data from Biopac DAQ, while computer B recorded the radar output data from NI DAQs.

TABLE I
SUMMARY OF BIOMETRIC INFO ON THE STUDY POPULATION

Metrics	Mean $\pm$ STD
Age	35.65 $\pm$ 13.42 [yr]
Weight	70.37 $\pm$ 12.27 [kg]
Height	169.65 $\pm$ 9.09 [cm]
BMI	24.4 $\pm$ 4.0
Max waist circumference	86.47 $\pm$ 10.9 [cm]
Min waist circumference	82.53 $\pm$ 10.5 [cm]
Max chest circumference	94.47 $\pm$ 7.6 [cm]
Min chest circumference	90.2 $\pm$ 6.4 [cm]
Max chest breath	30.93 $\pm$ 4.4 [cm]
Min chest breath	27.6 $\pm$ 4.6 [cm]
Max chest depth	23.57 $\pm$ 3.0 [cm]
Min chest depth	20.36 $\pm$ 1.5 [cm]

this study. After the demodulation, the threshold-based peak detection was used to identify the exact locations of respiration peaks and valleys for RSA assessment; the autocorrelation with statistical analysis was used to extract the heart rate information for both radar and ECG signals. This particular autocorrelation method incorporates mean and median of the physiological signal rates. The output of the autocorrelation is compared to the statistical variables. The advantage of this method is that the incorrect rate caused by motion artifacts or noise will be eliminated.

III. HUMAN TESTING PROTOCOL

The experiments were conducted according to the Committee on Human Studies (CHS) protocol number 14884. The protocol for this study was approved by the CHS of the University of Hawaii system. The data were taken from subjects in seated and supine positions. The subjects wore normal clothing and were instructed to breath normally during the measurements. The challenges in designing the test setup for this study were simultaneous measurement of reference and radar signals, human safety, and signal integrity. Fig. 3 shows the block diagram of the system setup. The reference signals: ECG, finger pulse, respiration signals from piezoelectric sensors on upper and lower torso were captured through the Biopac data acquisition system

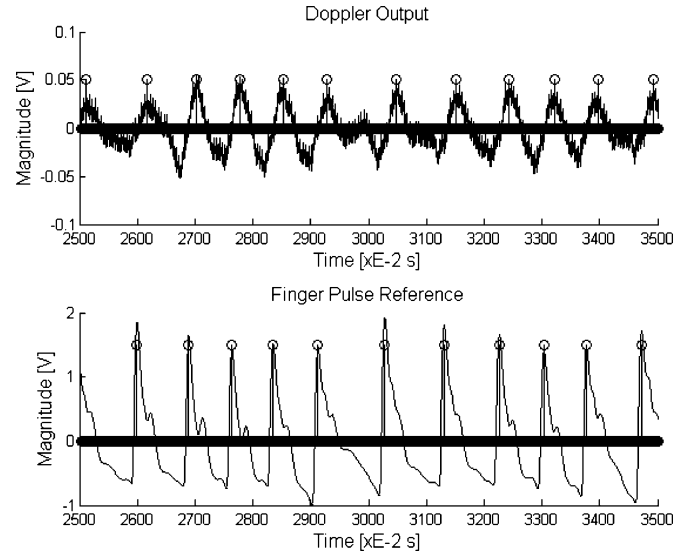


Fig. 4. 10-s sample data from the Doppler radar output after linear demodulation for heart signals (*top*), and finger pulse reference output (*bottom*) indicating all peaks that have been successfully detected. From [11].

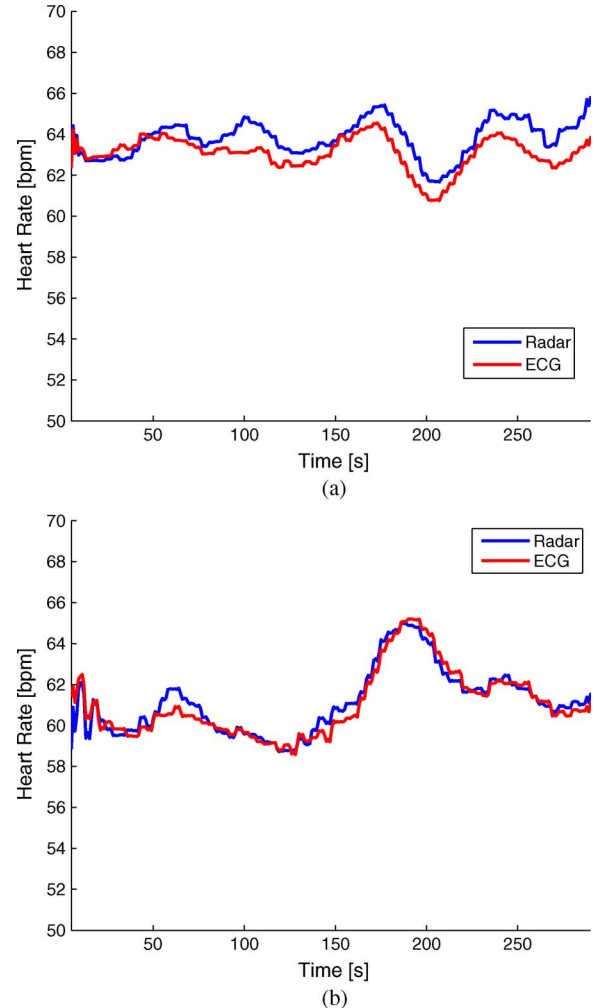


Fig. 5. Heart rate measurement in 5 min in: (a) seated and (b) supine positions of subject 2205. The radar output from supine position shows a better accuracy than from the seated position.

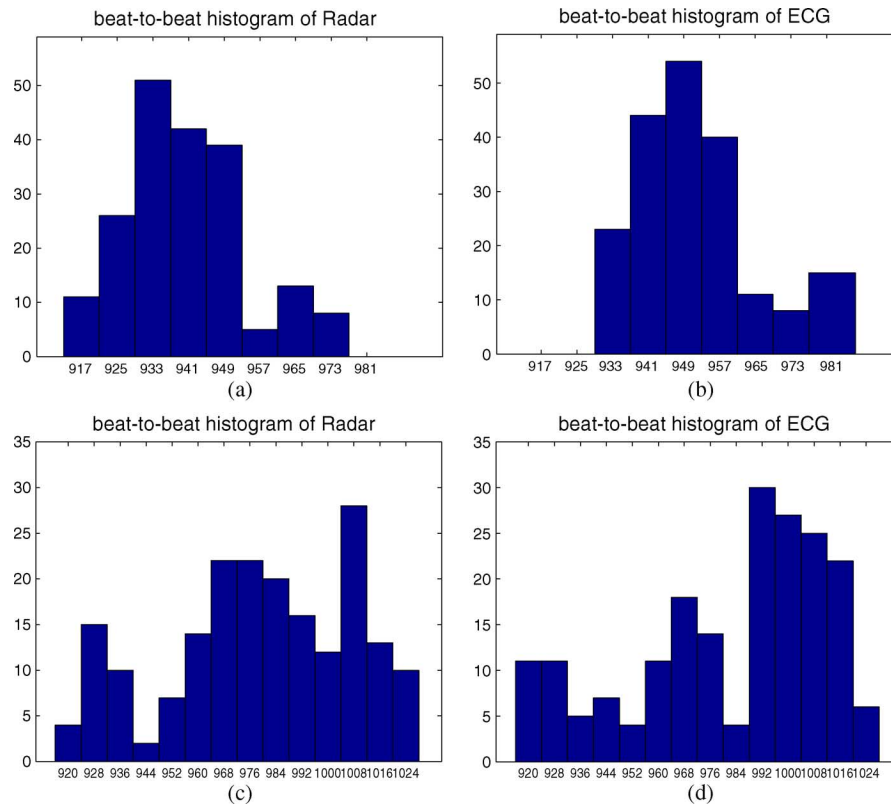


Fig. 6. Histogram of beat-to-beat intervals from subject 2205. (a) and (b) From seated position. (c) and (d) From supine position, which has wider distribution of beat-to-beat intervals. (a) Seated—radar. (b) Seated—ECG. (c) Supine—radar. (d) Supine—ECG.

(DAQ) [18]. The cardiopulmonary motion signals from Doppler radar were captured through the National Instrument DAQ.¹ The two DAQ systems captured data simultaneously; without a marker indication of the correct timing event, the data could not be analyzed correctly. Both DAQ systems were synchronized with a marker signal from the Biopac system via an optical isolator and a National Instrument DAQ card (National Instruments NI-DAQ PCI-6009). The first pulse was sent out when Biopac DAQ started capturing data. The pulse train had different width to minimize the synchronization error.

The Doppler radar operated at 2.4 GHz with 0-dBm power level at the antenna connector. The radar system used these following commercially available components: one transmitting antenna (Antenna Specialist ASPPT2988), one receiving antenna (Antenna Specialist ASPPT2988), two 0° power splitters (Mini-Circuits ZFC-2-2500), one 90° power splitter (Narda 4033C), and two mixers (Mini-Circuits ZFM-4212). The base-band output signals were amplified and filtered with low-noise amplifiers (Stanford Research Systems SR560) and then digitized with an on-board ADC of a National Instruments DAQ card (National Instruments NI-DAQ PCI-6259). The software to collect and process the data was written in MATLAB.

To evaluate the feasibility of HRV measurement via Doppler radar, three time-domain HRV indices, i.e.: 1) the SDNNs; 2) the root-mean-square of differences of successive beat-to-beat intervals (RMSSDs); and 3) the triangular index (HRV TI) were compared and observed for any differences. These three

time-domain indices were selected because they provide different HRV parameters. The SDNN provides an estimation of overall HRV. The RMSSD provides an estimation of HF components of HRV. The HRV TI provides the shape of the beat-to-beat interval distribution; uniform distributions representing large variability have large values and distributions with single large peaks have small values. A histogram with 8-ms bin was used in order to derive the HRV TI. The average heart rates and their Bland-Altman analysis were also observed in these experiments. The breath-by-breath basis RSA amplitudes were computed using the peak-valley estimation method, which utilizes both respiratory and heart period measurement data. When there were valid minimal and maximal heart periods for a breath, RSA was calculated as the difference between maximum and minimum heart periods in millisecond. The mean RSA value for a particular duration was the sum of the individual RSA scores for each breath divided by the number of breaths occurring during the measurement. High accuracy for HRV and RSA assessment was achieved.

IV. EXPERIMENTAL RESULTS

The results from 12 healthy volunteers are presented in this study. Table I summarized the characteristics of all subjects participated. In both seated and supine positions, the antennas were 1-m away from the front torso of the subjects. Fig. 4 shows a 10-s data from the Doppler radar output after linear demodulation for heart signal (*top*), and finger pulse reference (*bottom*), indicating all heart peaks that have been successfully detected.

¹[Online]. Available: <http://www.biopac.com>

TABLE II
AVERAGE HEART RATE WITH BLAND-ALTMAN ANALYSIS AND HRV INDICES IN SEATED POSITION FROM DOPPLER RADAR AND ECG

Subject	Avg. HR [BPM]		Bland-Altman		SDNN		RMSSD		HRV TI Index		HRV TI max bin	
	Radar	ECG	$\bar{d}$ [BPM]	S_d [BPM]	Radar	ECG	Radar	ECG	Radar	ECG	Radar	ECG
2202	67.58	70.27	2.6899	5.2314	75.23	17.41	601.24	49.70	6.96	4.88	852	852
2204	64.39	68.19	3.7963	2.6283	25.41	25.44	105.68	53.75	5.91	6.72	934	878
2205	63.88	63.15	-0.7328	0.5870	13.48	13.86	31.52	29.06	3.82	3.61	933	949
2206	69.79	72.80	3.0085	4.4347	32.80	13.93	258.78	24.22	8.48	4.53	823	815
2209	70.45	70.60	0.1561	2.7381	25.35	26.22	48.89	38.87	5.91	7.22	835	819
2210	63.31	63.18	-0.1290	2.0617	24.93	25.16	72.28	72.00	7.50	6.29	941	957
2302	68.67	71.08	2.4165	6.4347	55.14	20.11	482.36	47.59	3.82	3.98	873	841
2304	73.95	78.03	4.0752	7.5066	62.98	14.78	447.38	28.97	4.53	2.03	770	778
2306	63.96	58.66	5.2989	3.1783	33.83	33.14	139.76	53.93	7.80	6.72	932	1012
2401	68.04	68.32	0.2812	1.1528	18.90	18.93	71.92	65.94	4.33	4.43	908	908
2403	56.01	57.18	1.1662	1.8073	25.74	34.65	71.17	83.72	5.27	8.13	1063	1039
2404	64.63	60.95	3.6866	8.8792	88.43	17.67	569.50	39.81	6.96	5.91	971	995

TABLE III
AVERAGE HEART RATE WITH BLAND-ALTMAN ANALYSIS AND HRV INDICES IN SUPINE POSITION FROM DOPPLER RADAR AND ECG

Subject	Avg. HR [BPM]		Bland-Altman		SDNN		RMSSD		HRV TI Index		HRV TI max bin	
	Radar	ECG	$\bar{d}$ [BPM]	S_d [BPM]	Radar	ECG	Radar	ECG	Radar	ECG	Radar	ECG
2202	70.72	71.45	0.7384	0.9575	18.26	22.36	30.66	37.63	3.82	5.00	863	855
2204	58.26	57.44	-0.8160	1.3442	27.25	29.51	86.47	40.21	5.91	5.57	1049	1049
2205	61.17	61.13	-0.0411	0.7973	27.96	29.52	40.90	41.11	6.96	6.50	1008	992
2206	66.77	70.06	3.2985	2.8418	11.65	19.14	70.26	32.19	2.57	3.98	886	862
2209	60.02	61.80	1.7749	2.0546	22.28	18.43	80.52	34.33	6.29	3.25	1011	955
2210	64.32	64.17	-0.1468	0.4200	36.86	36.00	58.26	52.17	4.76	6.29	950	950
2302	66.65	67.97	1.3207	2.4439	24.47	15.84	67.37	22.94	5.91	4.64	882	866
2304	72.87	74.74	1.8718	1.3737	14.07	9.82	55.78	21.50	3.48	3.42	820	796
2306	62.81	62.66	-0.1517	1.5581	47.05	41.40	128.72	53.50	5.27	6.72	929	929
2401	58.59	58.54	-0.0479	0.9809	18.23	18.40	61.75	75.00	4.15	3.98	1029	1021
2403	55.07	55.97	0.9026	3.1365	31.38	38.55	95.90	91.59	6.29	6.29	1125	1125
2404	59.89	59.59	-0.3002	1.1920	12.52	18.33	67.84	39.28	4.31	4.56	1007	999

A. HRV

A 5-min window devoid of motion artifacts as much as possible was selected from each measurement based on visual inspection. The heart signal output from the linear-demodulated Doppler radar was then compared with the ECG reference signal for heart rate and HRV indices. The average heart rates and the Bland-Altman analysis, comparing the bias magnitude ($\bar{d}$) and the standard deviation (S_d) of the heart rates, were also observed. The examples of the heart-rate measurements in both seated and supine positions are shown in Fig. 5. In these experiments to assess the feasibility of HRV measurement via Doppler radar, the three time-domain HRV indices were compared and observed for any differences. These three indices are SDNN, RMSSD, and HRV TI. Each time the N beat occurs is denoted as $t(n)$ for $n \in \{1, \dots, N\}$. The interval between beats is denoted as $\delta(n) = t(n) - t(n-1)$. The SDNN is determined as

$$\text{SDNN} = \sqrt{\frac{1}{N-2} \sum_{n=2}^N (\delta(n) - \bar{\delta})^2} \quad (4)$$

where $\bar{\delta}$ is the average beat-to-beat interval. The RMSSD is determined as

$$\text{RMSSD} = \sqrt{\frac{1}{N-2} \sum_{n=3}^N [\delta(n) - \delta(n-1)]^2} \quad (5)$$

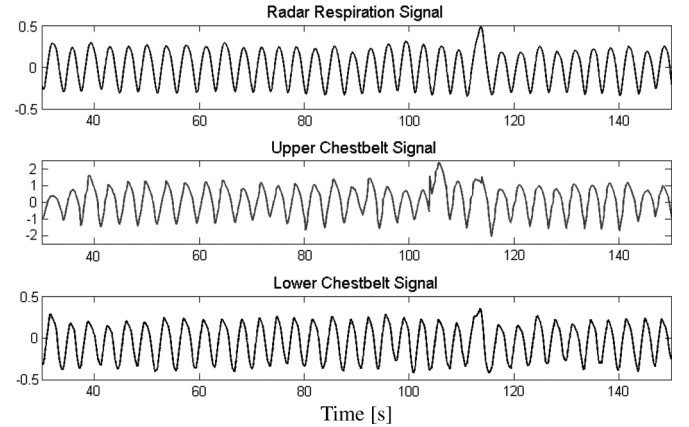


Fig. 7. Two-minute window of respiration signals from radar (top), upper chest belt (middle), and lower chest belt (bottom). This is an example of how the respiratory signal from the radar is more accurate than one of the chest-belt reference, which may have been contributed to improper placement of the belt sensor.

The HRV TI metric is defined in terms of a histogram of beat-to-beat intervals. $b(t_i)$ represents the number of intervals in the i th bin centered at t_i . The HRV TI is determined as

$$\text{HRVTI} = \frac{1}{\max_i b(t_i)} \sum_{i=1}^{N_b} b(t_i) = \frac{N-1}{\max_i b(t_i)} = \frac{N-1}{Y} \quad (6)$$

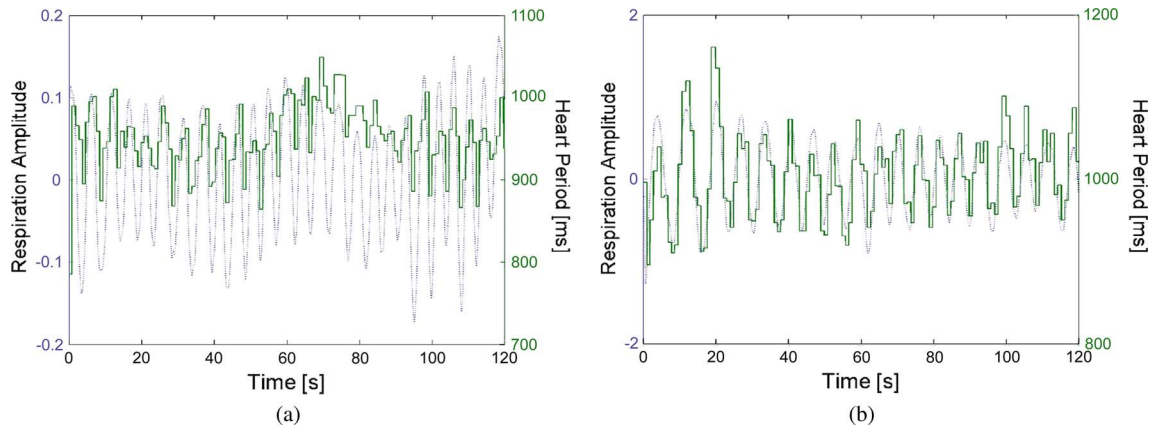


Fig. 8. 2-min sample data of the two-axis respiration sinus arrhythmia plots representing the respiratory signal (dotted line) and the heart period (full line) for a subject in: (a) seated position and (b) supine position. The heart periods were extracted from ECG signals, while the respiration patterns were extracted from Doppler radar signals. The cyclical heart rate varied in time, due to RSA, corresponded to the respiration signals.

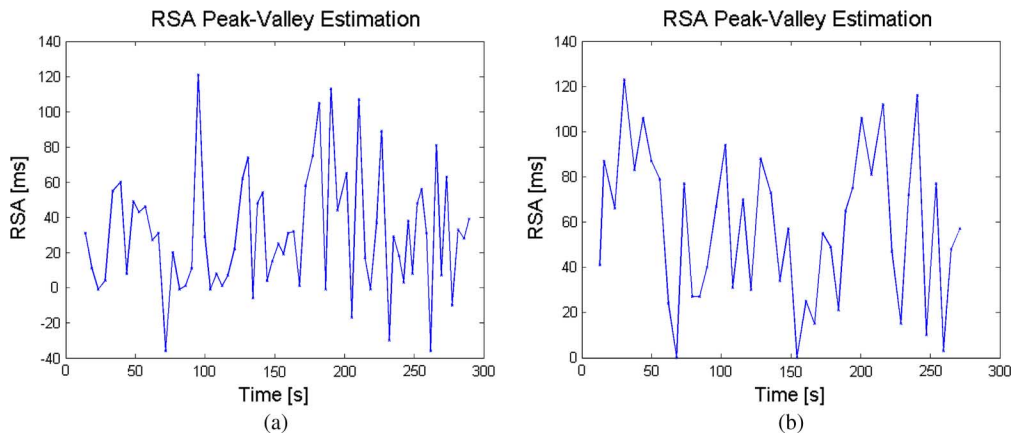


Fig. 9. RSA peak-valley amplitude estimation for: (a) seated position and (b) supine position for subject 2205.

The examples of the histograms of RR intervals extracted from Doppler radar and ECG signals are shown in Fig. 6. A histogram with an 8-ms bin was used in order to derive the HRV TI. Studies of autonomic nervous activity have shown that when the position was changed from upright seated to supine in a healthy individual, the percentage change in the mean RR interval and normalized HF power were increased [19], [20]. The same observation can be seen in this study. The heart rates in the supine position varied more than in the seated position. The summary of the heart rate with Bland-Altman analysis and HRV indices from Doppler radar and ECG are shown in the following tables: Table II for the results from subjects in a seated position, and Table III for the results from subjects in a supine position.

The average heart rates extracted from an ECG were similar to the average heart rates extracted from radar in both positions. However, Bland-Altman analysis reveals that the heart-rate measurements from the supine position were more accurate. Most of the bias magnitudes and the standard deviations in a supine position were smaller than those in a seated position for the same subjects.

The beat-to-beat intervals or instantaneous heart rates are the two main parameters for HRV analysis. Inaccurate measurement of both metrics will have a direct effect in HRV index calculation. The heart rates obtained using Doppler radar were extracted with high accuracy for 5-min data samples with minimum motion artifacts.

For supine position measurements, all SDNNs of Doppler radar and ECG differed less than ± 9 ms of each other, while all RMSSDs differed less than ± 76 ms. All maximum bins for HRV TIs were within ± 56 ms (seven bins).

For seated position measurements, the HRV indices were not quite as accurate. Five out of 12 subjects shown in Table I had SDNNs of Doppler radar that differed more than ± 9 ms. A different set of five subjects had RMSSDs that differed more than ± 76 ms. One subject had a maximum HRV TI bin that was not within ± 56 ms.

B. Respiration Sinus Arrhythmia

The RSA phenomenon is usually observed through the simultaneous measurement of heart rate and respiration. Current techniques for measuring respiration effort/movement typically require direct contact with the patient through various chest bands that may impede unrestricted chest motion. Fig. 7 shows an example of the error that may occur when using a chest-band technique versus Doppler radar. The result from upper chest belt did not agree with the Doppler radar and lower chest belt. Fig. 8 illustrates the two-axis RSA plot of heart rates and respiration patterns captured simultaneously. The heart periods were extracted from ECG signals, while the respiration patterns were extracted from Doppler radar signals. The cyclical heart rate variation in time, due to RSA, corresponded to the respiration signals. Examples of RSA peak-valley amplitudes from Doppler

TABLE IV
RSA PEAK-VALLEY ESTIMATION OF SEATED SUBJECTS

Subject	RSA Peak-Valley [ms]			Bland-Altman Analysis [ms]			
	Radar	Upper belt	Lower belt	Radar vs. Up d	Up S _d	Radar vs. Low d	Low S _d
2202	39.79	42.20	31.82	2.89	65.07	-7.76	64.50
2204	34.45	40.52	32.48	3.40	22.17	-4.73	104.6
2205	25.29	26.61	28.32	1.32	23.18	3.03	41.49
2206	31.83	32.79	30.29	1.68	110.3	-0.38	110.6
2209	26.45	13.55	26.14	-12.91	32.09	-0.32	29.74
2210	65.48	87.94	78.92	22.46	158.5	13.44	49.86
2302	31.69	44.31	25.83	12.61	86.26	-6.61	83.06
2304	57.41	59.24	48.61	-0.89	55.77	-10.47	59.15
2306	132.5	166.2	86.10	35.81	309.3	-46.05	301.4
2401	85.67	101.7	116.57	16.06	170.0	30.90	161.0
2403	36.92	41.57	42.51	4.65	26.58	5.60	25.66
2404	39.40	42.59	48.90	1.72	41.43	8.90	32.52

TABLE V
RSA PEAK-VALLEY ESTIMATION OF SUPINE SUBJECTS

Subject	RSA Peak-Valley [ms]			Bland-Altman Analysis [ms]			
	Radar	Upper belt	Lower belt	Radar vs. Up d	Up S _d	Radar vs. Low d	Low S _d
2202	43.56	66.18	49.37	22.62	146.4	6.94	149.6
2204	23.62	31.89	24.50	8.27	29.35	1.89	25.93
2205	112.7	85.19	114.7	-27.52	55.36	2.00	42.09
2206	19.57	16.28	22.11	-2.97	22.80	2.83	21.40
2209	35.61	57.03	39.41	22.48	76.52	4.55	70.20
2210	86.67	61.81	57.47	-24.86	167.2	-29.19	159.4
2302	30.63	29.60	29.90	-1.03	38.51	-0.73	19.46
2304	117.3	141.6	126.6	1.54	405.4	-1.27	392.9
2306	144.2	124.7	132.0	-18.76	423.4	-9.73	265.8
2401	209.7	165.8	133.6	-43.97	250.3	-75.38	72.22
2403	94.72	105.0	98.53	15.04	315.1	8.13	249.3
2404	30.16	21.08	22.43	-9.08	43.84	-7.74	41.54

radar are shown in Fig. 9. This method uses both respiratory and heart period measurement data. First, the heart-period time series were synchronized with the inspiration and expiration periods. The RSA magnitude was computed on a breath-by-breath basis. When there were valid minimal and maximal heart periods for a breath, RSA was calculated as the difference between maximum and minimum heart periods in milliseconds. The mean RSA value for a particular duration was the sum of the individual RSA score for each breath divided by the number of breaths occurring during the measurement. The RSA peak-valley indices from radar, upper chest belt, and lower chest belt are shown in Tables IV and V along with the Bland-Altman analysis of the peak-valley amplitudes from radar against the two chest belts.

The same 5-min duration of data were used to assess the feasibility of RSA measurement. In the subjects with normal RSA, the results from the radar correlated well with both upper and lower chest belts. These subjects had, on the average, the Bland-Altman bias magnitude of 7 ms with a standard deviation of 51 ms for seated positions and bias magnitude of 10 ms with a standard deviation of 48 ms for supine positions. For the sub-

jects with high RSA indices in both seated and supine positions, the RSA peak valley from the Doppler radar did not correlate as well with the chest belts. The Bland-Altman analysis shows that the subjects with standard deviation of 100 ms or higher tended to be those with high RSA.

V. CONCLUSION

The results presented in this paper have shown that using a quadrature Doppler radar system with a linear demodulation method provides fairly accurate analysis for HRV and RSA. The HRV assessment from Doppler radar had a good correlation to the reference when the subjects were in a supine position. It was also found, as predicted, that supine subjects had a higher degree of HRV. Furthermore if the subjects had normal RSA regardless of the positions, the results correlated better to the reference. While this remote physiological sensing technique is currently limited to sedentary and supine patients, due to difficulty with extracting cardiopulmonary signals in the presence of gross subject motion, the capability to extract respiration and heart signals independently provides a powerful tool for future health monitoring. This technique may be particularly suitable for sleep apnea monitoring, infant SIDS monitoring, fatigue monitoring, in-hospital monitoring, and home health care.

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REFERENCES

- [1] H. Stauss, "Heart rate variability," *Amer. J. Physiol.*, vol. 285, no. 5, pp. R927-R931, 2003.
- [2] P. Ivanov, L. Nunes Amaral, A. Goldberger, S. Havlin, M. Rosenblum, Z. Struzik, and H. E. Stanley, "Multifractality in human heartbeat dynamics," *Nature*, vol. 399, pp. 461-465, Jun. 1999.
- [3] Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology, "Heart rate variability: Standards of measurement, physiological interpretation and clinical use," *Circulation*, vol. 93, pp. 1043-1065, 1996.
- [4] R. Wood, B. Maraj, C. M. Lee, and R. Reyes, "Short-term heart rate variability during a cognitive challenge in young and older adults," *Age Aging*, vol. 31, pp. 131-135, 2002.
- [5] M. F. Hilton *et al.*, "Frequency spectral analysis of heart rate variability as a diagnostic marker of the sleep apnea syndrome," *Med. Biol. Eng. Comput.*, vol. 37, no. 6, pp. 760-769, Nov. 1999.
- [6] K. C. Bilchick *et al.*, "Prognostic value of heart rate variability in chronic congestive heart failure (veterans affairs' survival trial of antiarrhythmic therapy in congestive heart failure)," *Amer. J. Cardiol.*, vol. 90, pp. 24-28, Jul. 2002.
- [7] M. Ikawa *et al.*, "Effects of combination psychotropic drug treatment on HRV in psychiatric patients," *Psych. Clin. Neurosci.*, vol. 55, no. 4, pp. 341-345, 2001.
- [8] P. M. Macabe, B. G. Yongue, P. K. Ackles, and S. W. Porges, "Changes in heart period, heart-period variability, and a spectral analysis estimate of respiratory sinus arrhythmia in response to pharmacological manipulations of the baroreceptor reflex in cats," *Psychophysiology*, vol. 22, no. 2, pp. 195-203, 1985.
- [9] W. Massagram, N. Hafner, S. Yamada, O. Boric-Lubecke, and V. M. Lubecke, "Feasibility of HRV measurements with single-channel Doppler radar," in *IEEE Antenna Propag. Symp. Dig.*, Honolulu, HI, Jun. 2007, pp. 269-272.
- [10] W. Massagram, N. Hafner, B.-K. Park, O. Boric-Lubecke, A. Host-Madsen, and V. Lubecke, "Feasibility of heart rate variability measurement from quadrature Doppler radar using arctangent demodulation with DC offset compensation," in *IEEE 29th Int. Eng. Med. Biol. Soc. Conf.*, Lyon, France, Sep. 2007, pp. 1643-1646.

- [11] O. Boric-Lubecke, W. Massagram, V. M. Lubecke, A. Høst-Madsen, and B. Jokanovic, "Heart rate variability assessment using Doppler radar with linear demodulation," in *IEEE MTT-S Eur. Microw. Week*, Amsterdam, The Netherlands, Oct. 2008, pp. 420–423.
- [12] J. C. Lin, "Microwave sensing of physiological movement and volume change: A review," *Bioelectromagnetics*, vol. 13, pp. 557–565, 1992.
- [13] T. Kondo, T. Uhlig, P. Pemberton, and P. D. Sly, "Laser monitoring of chest wall displacement," *Eur. Resp. J.*, vol. 10, pp. 1865–1869, 1997.
- [14] G. Ramachandran, S. Swarnamani, and M. Singh, "Reconstruction of out-of-plane cardiac displacement patterns as observed on the chest wall during various phases of ECG by capacitance transducer," *IEEE Trans. Biomed. Eng.*, vol. 38, no. 4, pp. 383–385, Apr. 1991.
- [15] B. K. Park, S. Yamada, O. Boric-Lubecke, and V. M. Lubecke, "Single-channel receiver limitations in Doppler radar measurements of periodic motion," presented at the IEEE RWS Student Paper Competition, San Diego, CA, Jan. 2006.
- [16] B. K. Park, V. M. Lubecke, and O. Boric-Lubecke, "Arctangent demodulation in quadrature Doppler radar receiver system with DC offset compensation," *IEEE Trans. Microw. Theory Tech.*, vol. 55, no. 5, pp. 1073–1079, May 2007.
- [17] B.-K. Park, V. M. Lubecke, and O. Boric-Lubecke, "Center tracking quadrature demodulation for a Doppler radar motion detector," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2007, pp. 1323–1326.
- [18] B. K. Park, A. Vergara, O. Boric-Lubecke, V. M. Lubecke, and A. Host-Madsen, "Center tracking quadrature demodulation for a Doppler radar motion detector," *IEEE Trans. Microw. Theory Tech.* [Online]. Available: <http://www.ee.hawaii.edu/~madsen/papers/index.html>, submitted for publication
- [19] G. Y. Chen, C. D. Kuo, M. J. Yang, H. M. Lo, and Y. S. Tsai, "Comparison of supine and upright positions on autonomic nervous activity in late pregnancy: The role of aortic caval compression," *Anaesthesia*, vol. 54, no. 3, pp. 215–219, Mar. 1999.
- [20] A. Malliani *et al.*, "Individual recognition by heart rate variability of two different autonomic profiles related to posture," *Circulation*, vol. 96, no. 12, pp. 4143–4145, Dec. 1997.



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